

Lab Five

ID1063: Introduction to Programming

1. Write a program that accepts a positive integer n and prints an $n \times n$ multiplication table.

For example, if $n = 4$, the output should be:

```
1  2  3  4
2  4  6  8
3  6  9 12
4  8 12 16
```

2. Write a program to accept a positive integer n , two vectors of length n into two arrays, and find their dot product. Example run:

```
Enter the value of n: 5
```

```
Enter the values of the first vector: 3 -1 10 2.5 6
```

```
Enter the values of the second vector: 0.5 4 1 0 -0.5
```

```
The dot product is 4.5.
```

3. Write a program that accepts a positive integer n , followed by n integers, each between 0 and 9, and stores the values in an array. Print the number of occurrences of each integer from 0 to 9.

For example:

```
Enter the value of n: 12
```

```
Enter the values: 3 1 4 1 5 9 2 6 5 3 5 1
```

```
0: 0
```

```
1: 3
```

```
2: 1
```

```
3: 2
```

```
4: 1
```

```
5: 3
```

```
6: 1
```

```
7: 0
8: 0
9: 1
```

4. Write a program to accept two integers m and n and display the perimeter of a $m \times n$ rectangle.

Example run:

Enter the value of m : 4

Enter the value of n : 7

```
* * * * * * *
*               *
*               *
* * * * * * *
```

5. Write a program that accepts a positive integer n , followed by n floating-point values, and stores the values in an array.

Compute the average of the values and then print the number of values that are strictly greater than the average.

For example:

```
Enter the value of n: 6
Enter the values: 4.5 2.0 7.5 5.0 3.0 8.0
Average: 5.0
Number of values above the average: 2
```

6. Write a program that accepts a positive integer n , followed by n integers stored in an array. Determine whether there are two distinct positions in the array containing the same value.

If there are two such positions, print one such pair and the corresponding common value. Otherwise, print that the array has no repeated values.

For example:

```
Enter the value of n: 6
Enter the values: 4 7 2 9 7 5
Repeated value: 7, Positions: 1, 4
```

Another example:

Enter the value of n: 5
Enter the values: 4 7 2 9 5
There are no repeated values.

Remark: Array positions in C start from zero.